

WHAT HAPPENS WHEN E-COMMERCE REVENUE JUMPS +137% YOY?

Behind Target's 99,000+ orders across 27 Brazilian states:
An Operational & Regional Logistics SQL Deep Dive.

99,441

Delivered Orders Analyzed

Sept 2016 – Aug 2018 span

27 States

4,119 Distinct Cities

Nationwide coverage map

3.5x Gap

Delivery Lead Time Gap

29.3 days (RR) vs 8.3 days (SP)

+137.38%

YoY Revenue Surge

Jan–Aug 2017 vs 2018 pacing

▣ WHY RECRUITERS & PEERS SHOULD CARE:

- Rapid revenue growth often conceals deep operational bottlenecks.
- Using Google BigQuery, CTEs & Window Functions, I uncovered why distant territories wait nearly a month for orders and pay 2.8x freight.
- Here is the step-by-step breakdown of SQL logic, findings & strategies.

THE BUSINESS PROBLEM

Scaling Across 4,119 Cities

Target experienced rapid growth across Brazil between 2016 and 2018. However, managing orders across a geographically vast continent presents acute supply chain, freight economics, and customer retention challenges.

1. Logistics Disparities

Are distant northern states suffering from unacceptable fulfillment delays?

2. Freight Equity

Are shipping charges discouraging conversions outside southeastern hubs?

3. Revenue Velocity

Is the top-line growth pacing sustainable across comparable quarters?

4. Consumer Habits

When do customers order, and how heavily do they rely on installment credit?

📊 GOOGLE BIGQUERY RELATIONAL ARCHITECTURE (6 TABLES):

- **orders:** 99,441 rows — order timestamps, status, carrier & estimated delivery dates
- **customers:** 99,441 rows — customer location keys across 27 Brazilian states
- **order_items:** 112,650 rows — item prices (BRL) and individual freight costs
- **payments:** 103,886 rows — payment type breakdown & installment frequencies
- **products & sellers:** 36,000+ rows — product dimensions, seller locations, and logistics

FINDING #1: FULFILLMENT VELOCITY

A 3.5x Regional Lead-Time Gap

Comparing average delivery lead time between the best & worst performing states:

FASTEST STATE: SÃO PAULO (SP)

8.3 Days

Average Delivery Time from purchase to doorstep.
Consistently beats estimated delivery dates by 11+ days.

SLOWEST STATE: RORAIMA (RR)

29.3 Days

Average Delivery Time — almost 1 full month waiting time!
~3.5x longer than orders delivered in São Paulo.

TOP 5 HIGHEST vs TOP 5 LOWEST DELIVERY DURATION STATES

State	Avg Delivery (Days)	State	Avg Delivery (Days)
RR (Roraima)	29.3 days	SP (São Paulo)	8.3 days
AP (Amapá)	27.2 days	PR (Paraná)	11.6 days
AM (Amazonas)	26.0 days	MG (Minas Gerais)	11.6 days
PA (Pará)	23.4 days	SC (Santa Catarina)	14.5 days
MA (Maranhão)	21.1 days	DF (Distrito Federal)	12.6 days

OPERATIONAL TAKEAWAY:

Northern territories suffer because logistics routes originate almost entirely from southeastern warehouses. Target cannot scale nationwide without local hubs.

QUERY ARCHITECTURE

Window Ranking with DENSE_RANK()

How I isolated the top 5 highest and lowest lead-time states without breaking rank sequence:

```
● ● ● BIGQUERY SQL • DENSE_RANK() FOR REGIONAL EXTREMES

-- 1. Calculate actual delivery duration per delivered order
WITH dt AS (
  SELECT o.order_id, c.customer_state,
         DATE_DIFF(order_delivered_customer_date,
                   order_purchase_timestamp, DAY) AS time_to_deliver
  FROM `target_sql.orders` o
  JOIN `target_sql.customers` c USING (customer_id)
  WHERE order_delivered_customer_date IS NOT NULL
),
-- 2. Compute state-level averages and window ranking
tr AS (
  SELECT customer_state,
         ROUND(AVG(time_to_deliver), 2) AS avg_delivery_time,
         DENSE_RANK() OVER (ORDER BY AVG(time_to_deliver) DESC) AS high_rank,
         DENSE_RANK() OVER (ORDER BY AVG(time_to_deliver)) AS low_rank
  FROM dt
  GROUP BY customer_state
)
-- 3. Extract top 5 extremes dynamically
SELECT customer_state, avg_delivery_time,
       CASE WHEN high_rank <= 5 THEN 'top 5 highest'
            WHEN low_rank <= 5 THEN 'top 5 lowest' END AS tier
FROM tr WHERE high_rank <= 5 OR low_rank <= 5
ORDER BY avg_delivery_time DESC;
```

❑ WHY DENSE_RANK() INSTEAD OF ROW_NUMBER() OR RANK()?

- Handles Ties Cleanly: States with identical average delivery times share the same rank.
- No Skipped Positions: Consecutive ranks remain intact (1, 2, 3... rather than 1, 1, 3).
- Dual Window Strategy: Generates both high and low rankings in a single database pass.

FINDING #2: FREIGHT DISPARITY

The Hidden Penalty on Remote States

Customers in distant territories don't just wait longer—they pay nearly 3x more for shipping:

SP (São Paulo)**17.37 BRL**

Lowest Avg Freight across Brazil

RR (Roraima)**48.59 BRL**

2.8x higher than São Paulo!

▣ WHY FREIGHT DISPARITY CREATES CHECKOUT ABANDONMENT:**Seller Clustering in Southeast:**

Over 70% of active sellers and fulfillment centers are located in SP, PR, and RJ. Long-haul transport drives up logistics rates exponentially.

The 'Distance Tax':

For a standard 100 BRL basket, an extra 48 BRL freight charge represents a nearly 50% checkout surcharge on the customer!

Depressed Northern Demand:

States like RR, AP, and AC have orders barely reaching double digits—not due to lack of interest, but because shipping cost is prohibitive.

▣ THE STRATEGIC INTERVENTION:**Target cannot rely purely on digital marketing in northern Brazil.**

Marketing ad spend is wasted if customers drop out at the final payment step due to high freight. Subsidized freight thresholds or regional seller onboarding is required to unlock demand.

FINDING #3: ORDER BEHAVIOR

When & How Brazilian Consumers Buy

Analyzing ordering hours and payment preferences reveals two vital operational patterns:

1. PEAK ORDERING WINDOW: AFTERNOON (13:00 – 18:00)

Hourly Breakdown:

Afternoon (13:00 - 18:00):	38,135 Orders	— PEAK TRAFFIC (Avg 14:00 - 15:00)
Night (19:00 - 23:00):	33,518 Orders	— High evening purchasing activity
Morning (07:00 - 12:00):	23,359 Orders	— Gradual ramp-up during workday
Dawn (00:00 - 06:00):	4,429 Orders	— Minimal activity window

Operational Implication: Schedule cloud database backups and system maintenance strictly during Dawn (02:00 - 05:00) to ensure zero friction during afternoon peaks.

2. PAYMENT METHODS & MULTI-INSTALLMENT RELIANCE

Credit Card is the undisputed leader (>75% of total orders), followed by Boleto.

Installment Distribution Analysis:

- 1 Installment: 52,580 orders (50.6%) — Dominant for low-to-mid ticket items
- 2 – 4 Installments: 27,713 orders (26.7%) — Popular for mid-range categories
- 5 – 10 Installments: 22,544 orders (21.7%) — Crucial for electronics and high-AOV items
- > 10 Installments: 1,049 orders (1.0%) — High-ticket special purchases

Commercial Insight: Nearly 50% of orders rely on installment financing. Promoting zero-cost EMI / installment discounts directly drives cart conversions.

QUERY ARCHITECTURE

Measuring YoY Growth with LAG()

Comparing identical 8-month periods (Jan–Aug) in 2017 vs 2018 to eliminate seasonality bias:

 BIGQUERY SQL • YoY REVENUE PACING VIA LAG()

```
-- 1. Aggregate total cost for Jan-Aug in 2017 and 2018
WITH tcs AS (
  SELECT
    EXTRACT(YEAR FROM o.order_purchase_timestamp) AS order_year,
    ROUND(SUM(p.payment_value), 2) AS order_total_cost
  FROM `target_sql.orders` o
  JOIN `target_sql.payments` p ON o.order_id = p.order_id
  WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018)
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
  GROUP BY EXTRACT(YEAR FROM o.order_purchase_timestamp)
),
-- 2. Window LAG to access previous year revenue in same row
lags AS (
  SELECT order_year, order_total_cost,
    LAG(order_total_cost) OVER (ORDER BY order_year) AS pre_year_cost
  FROM tcs
)
-- 3. Compute exact percentage growth rate
SELECT order_year, order_total_cost, pre_year_cost,
  ROUND((order_total_cost - pre_year_cost)/pre_year_cost * 100, 2)
  AS percentage_growth
FROM lags ORDER BY order_year;
```

RESULT SET FROM BIGQUERY EXECUTION:

- 2017 (Jan–Aug): 3,707,750.32 BRL
- 2018 (Jan–Aug): 8,801,634.64 BRL

☑ **VERIFIED YoY GROWTH: +137.38%**

BUSINESS STRATEGY

4 Data-Backed Action Plans

Translating data insights into high-ROI operational and business decisions:

1. Establish Northern Regional Micro-Hubs

Partner with localized 3PL providers in Manaus / Belém to slash the 29-day delivery cycle in RR and AM down to under 12 days, reducing cancellation rates.

2. Implement Tiered Freight Subsidies

Introduce free shipping thresholds (e.g. Orders > 150 BRL) for northern states to eliminate the 48 BRL checkout barrier and unlock dormant markets.

3. Provision Infrastructure for Afternoon Peaks

Autoscale database resources and customer support between 13:00 and 18:00 when ~40% of daily transactions occur; shift backups to 03:00 dawn.

4. Expand Zero-Cost Installment Programs

With ~50% of orders relying on installment financing, offer zero-fee EMI promotions on high-AOV categories to increase average cart size.

CAPABILITY SHOWCASE

Skills & Technologies Applied

How my analytical background was leveraged throughout this case study:

Google BigQuery

Cloud Relational Querying

Multi-table relational schema exploration, column validation, large table joins (100K+ records).

CTEs, DENSE_RANK(), LAG()

Advanced SQL Patterns

Common Table Expressions, window ranking for state extremes, lag pacing for YoY financial metrics.

Delivery SLA & Freight Modeling

Supply Chain & Logistics

Measuring actual vs estimated transit days, identifying shipping cost inflation by territory.

Ex-BPCL Operations Grounding

Applied Operational Acumen

1 year monitoring retail sales & stock across 350+ petrol stations applied directly to retail logistics.

DIVE DEEPER

Explore the Code & Case Study

Everything in this case study is open-source and documented in detail:

📁 GitHub Repository

github.com/Hazardous9hub/Target-SQL-Business-Case-Study

Complete SQL scripts, schema documentation, and 39 BigQuery query screenshots.

🌐 Live Portfolio Website

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Interactive case study deep dives, project demos, resume, and credentials.

👤 LinkedIn Profile

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Connect with me for data analytics discussions, problem solving, and opportunities.

👤 SHIVALING BATTARKI

Data Analyst | Ex-BPCL | Scaler DSML Fellow | B.E. Mechanical (8.67 CGPA)

- 1 Year on-ground operations analytics at BPCL (350+ petrol stations, SAP ECC & MIS).
- HackerRank 5-Star Gold SQL | Scaler DSML Fellow | BigQuery & Tableau Specialist.

📩 **Actively seeking Data Analyst & Business Analyst roles. Let's talk!**